International Islamic University Chittagong
Department of Computer Science & Engineering
B.Sc. in CSE *Online Mid Term Assessment* **, Autumn -2024**
Course Title: Mathematics-III Course Code: MATH-2307 (New)
Course Title: Mathematics-IV Course Code: MATH-2401 (Old)

Total Marks: 15 Section: 3AM Submission Time: 04 hours

Answer all the questions; *Be sure to follow the instructions below* for preparing and submitting the assignment;Figures in the right hand margin indicate full marks

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1. a) Check whether it is singular matrix $\begin{bmatrix} 1 & 2 & f \\ 2 & 4 & e \\ 3 & 5 & d \end{bmatrix}$; Where d , e and f are the digits of your own metric number 2

b) Find the inverse of $A = \begin{bmatrix} 1 & f+1 \\ 3 & e \end{bmatrix}$ Where e and f are the digits of your own metric number 3

c) Construct the 4×4 Matrix A having: 2

$$\begin{aligned} \mathbf{A} = (\mathbf{a}_{ij}) &= \mathbf{3i + j} \quad ;\mathbf{when\ i > j} \\ &= \mathbf{0} \quad ;\mathbf{when\ i = j} \\ &= \mathbf{2i - j} \quad ;\mathbf{when\ i < j} \end{aligned}$$

2. a) Determine which of the following vectors are Eigen vectors for $A = \begin{pmatrix} 2 & f \\ -1 & -6 \end{pmatrix}$ showing your analysis procedure graphically 3

(a)$\begin{pmatrix} a \\ e \end{pmatrix}$ (b)$\begin{pmatrix} c \\ d \end{pmatrix}$ (c)$\begin{pmatrix} b \\ f \end{pmatrix}$; Where a , b, c, d, e and f are the digits of your own metric number

b) Which variable(s) is (are) free variables for the linear system of equations 1

$$7x + 5y + 9z = 6$$
$$w = 1$$

3. a) Test whether $\lambda_1 = e$ and $\lambda_2 = f$ are eigen values for $A = \begin{bmatrix} 8 & 3 \\ 2 & 7 \end{bmatrix}$; Where e and f are the digits of your own metric number 1

b) Solve the system of linear equations 3

$$\begin{aligned} (f+1)x + 9y - z &= 27 \\ x - 8y + 16z &= 10 \\ ax + y + 15z &= 37 \end{aligned}$$

Where a and f are the digits of your own metric number

Instructions

If your matric no is C171023 then you can assign your digit as follows

C	1	7	1	0	2	3
	a	b	c	d	e	f

To solve the above problem, you need to use your own metric number where needed